

Amazing Speech-Controlled SUDOKU SOLVING ROBOT

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For most people solving sudoku puzzles is not a particularly simple task. A novice may take many hours to finish a straightforward sudoku puzzle. Because of this, we wondered if it was possible to create a robot that could simply

by looking at a sudoku problem, regardless of how difficult it was, solve it within a few seconds?

Here, we look at how to create such straightforward sudoku solver robot, that, based on our voice commands, can quickly solve any sudoku puzzle for us. Also, it will show us the whole procedure of solving the puzzle. The robot can also talk and exhibit its own expressions to make it even more interesting.

Before proceeding with this robot's design, you need to understand the rules of sudoku. So, let's go through the rules before building the robot to solve the puzzles for us.

As shown in Fig. 1, a 9×9 grid is used to play sudoku. There are nine

large squares in all, each having a 3×3 grid of small boxes. You need to fill all the small boxes with numbers 1 through 9, without repeating any number in any of the rows or columns. In each sudoku puzzle, some of the boxes are already filled with the numbers.

Understanding a problem's input-output relationship is necessary before it can be solved. In this case, the image of a sudoku puzzle printed on any white paper or a digital sudoku puzzle on any screen will serve as our input and captured by our robot. It can be the picture of a sudoku puzzle from a newspaper that is shown in Fig. 1. The solved puzzle will be shown on our robot's screen as the output, as shown in Fig. 2.

Before building the robot, we



Fig. 1: Sample unsolved sudoku puzzle

5	3	9	1	7	2	6	8	4
4	7	8	9	6	3	1	5	2
2	6	1	5	8	4	7	9	3
8	9	7	6	4	5	3	2	1
1	2	4	8	3	9	5	7	6
3	5	6	7	2	1	8	4	9
9	8	5	4	1	6	2	3	7
6	4	2	3	5	7	9	1	8
7	1	3	2	9	8	4	6	5

Fig. 2: Sample solved sudoku puzzle

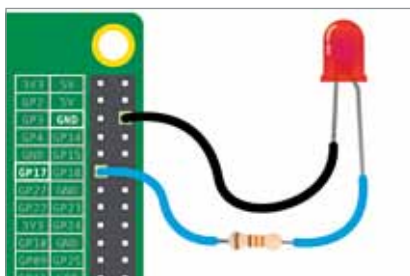


Fig. 3: LED connected with Raspberry Pi

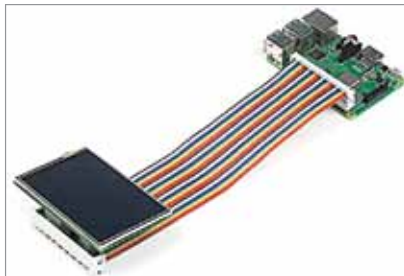


Fig. 4: Touch-screen connected with Raspberry Pi

Bill of Material	
Components	Quantity
Raspberry Pi 4 B/Raspberry Pi 3 B+ (Raspberry Pi 4 recommended)	1
Raspberry Pi camera module V2	1
Raspberry Pi touch-screen display	1
5W speaker	1
LED	1
5V, 2A power adaptor	1
USB microphone	1
Jumper wires, female - female	40
A robot toy body/case	1



Fig. 5: Raspberry Pi camera attached to the robot's body